

RRC-Raptor-V4H

- Connector

V0.0.3

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i. VERSION LIST

Description	Version	Date	Author
Connector 、 Pin define	V0.0.1	2024/09/12	KevinLin
Modify 4.8(Page13) 、 4.9(Page14) 、 4.10(Page15)	V0.0.2	2024/10/11	KevinLin
File attribute changes	V0.0.3	2025/01/22	kevinlin

ii. HASHTAG

#V4H

#Switch

#connector

1. Overview

Target :

Introducing the connector and pin definitions in Raptor-V4H.

2. Configuration Diagram

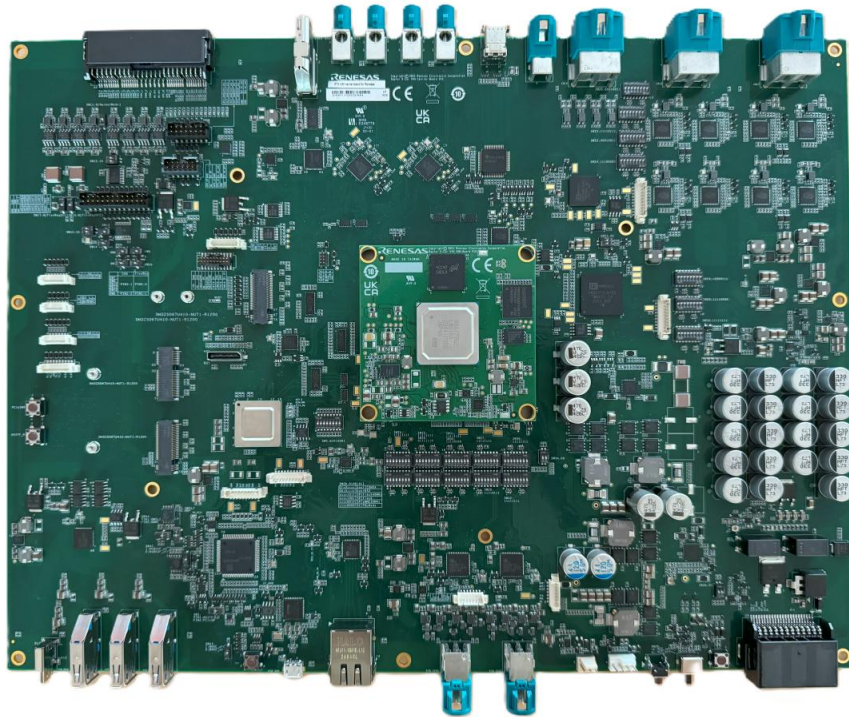


Figure 1. Top View

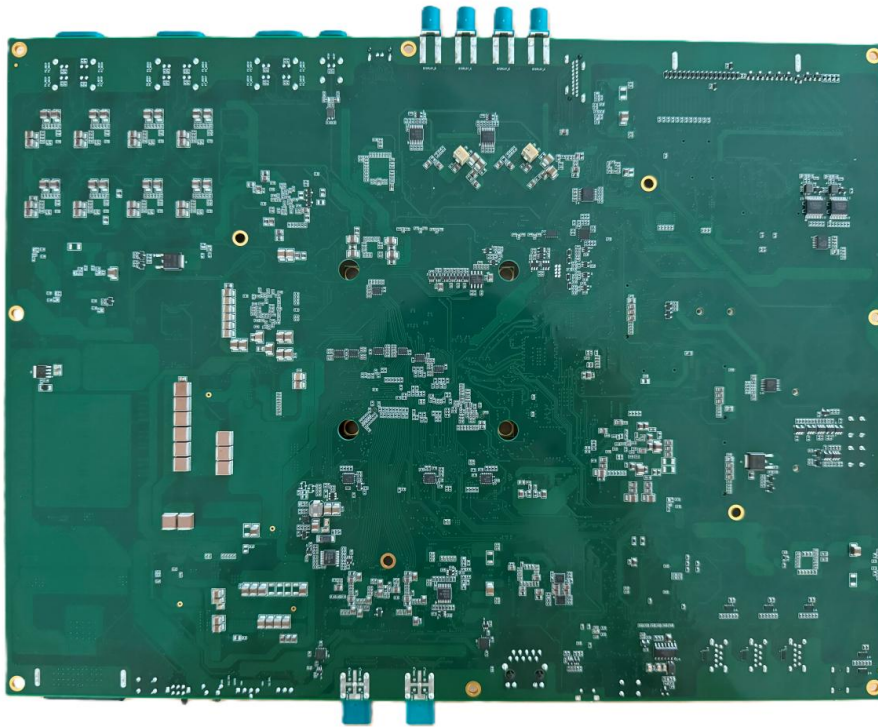


Figure 2. Bottom View

3. External Connector

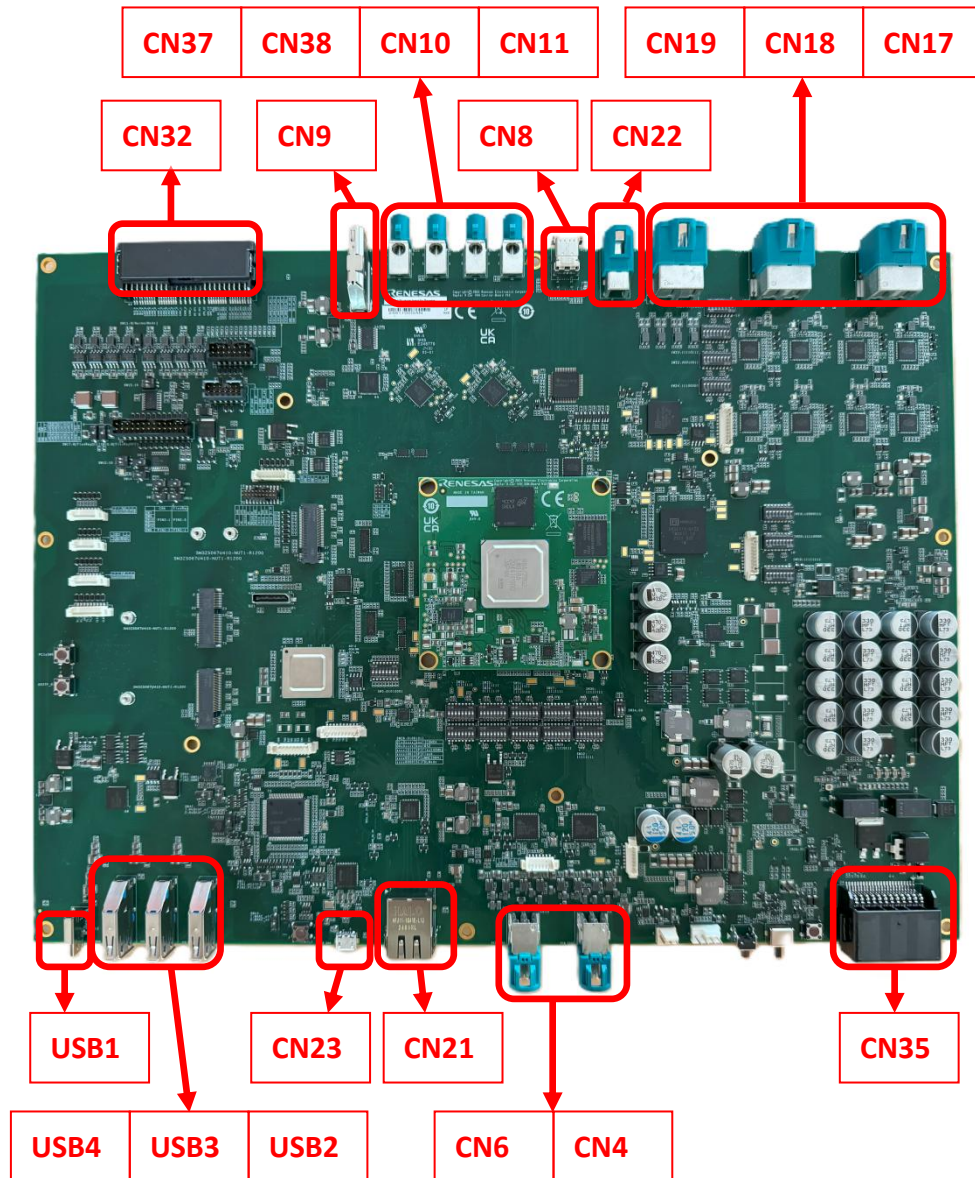


Figure 3. External Connector

4. Internal Connector

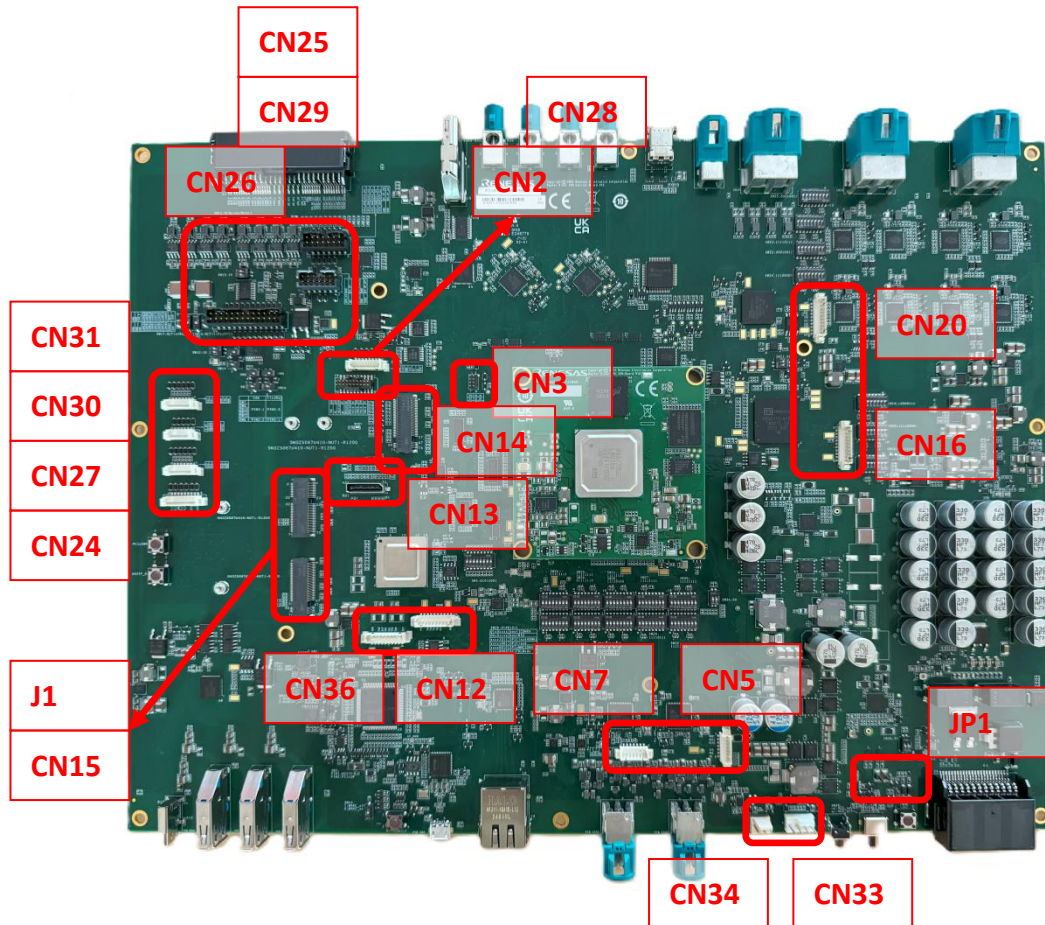
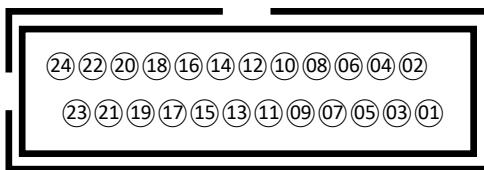


Figure 4. Internal Connector

4.1. CN35

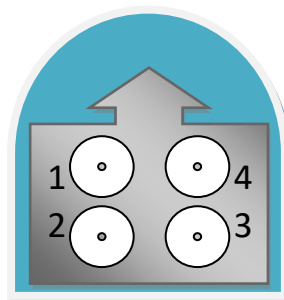
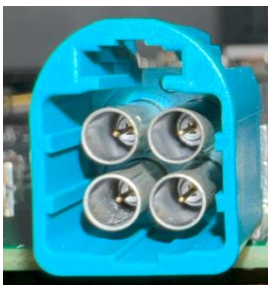
[Angle Pin Header (SMT)]



Function	PIN	PIN	Function
SG	H1	H2	SG
B -	1	2	B -
B -	3	4	B -
B -	5	6	B -
B -	7	8	B -
B +	9	10	B +
B +	11	12	B +
B +	13	14	B +
B +	15	16	B +
B +	17	18	B +
REFERENCE_LIN_SW	19	20	DR_DOOR_SW
IG1_CN	21	22	AS_DOOR_SW
CAN_L	23	24	CAN_H
SG	H3	H4	SG
SG	H5	H6	SG

4.2. CN4

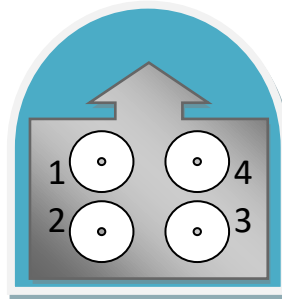
Camera to [CSI0] FAKRA



PIN	CN4	[U8]MAX96712	[U7]MAX20087
1	1	GMSL_CAM0_C	OUT_3
2	2	GMSL_CAM0_D	OUT_4
3	3	GMSL_CAM0_A	OUT_1
4	4	GMSL_CAM0_B	OUT_2

4.3. CN6

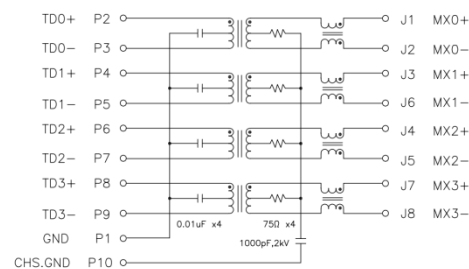
Camera to [CSI1] FAKRA



PIN	CN4	[U11]MAX96712	[U10]MAX20087
1	1	GMSL_CAM1_C	OUT_3
2	2	GMSL_CAM1_D	OUT_4
3	3	GMSL_CAM1_A	OUT_1
4	4	GMSL_CAM1_B	OUT_2

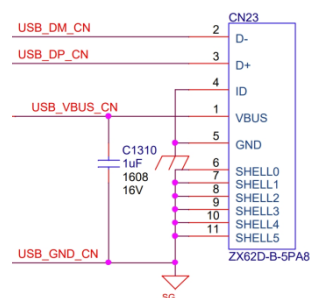
4.4. CN21

[Gigabit (10/100/1000) Ethernet RJ45] RJ45 with integrated magnetic



4.5. CN23

[USB microAB]



4.6. USB2 USB3 USB4

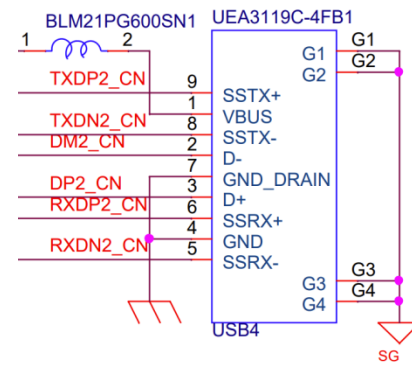
[USB TYPE A] USB3.0



USB4

USB3

USB2



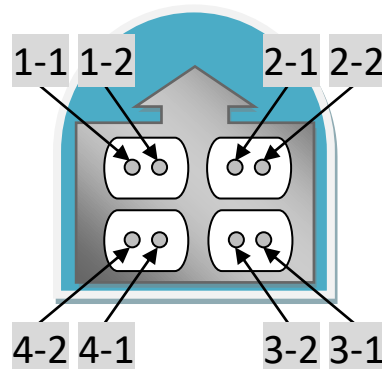
4.7. USB1

[USB TYPE C] USB3.0



4.8. CN17

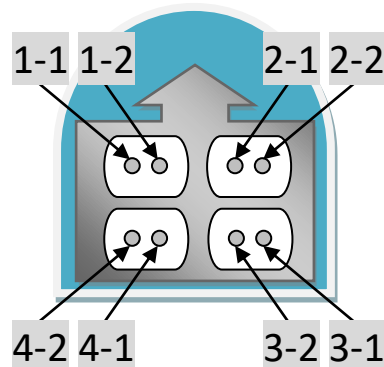
[410AF08C6TN1T1CZ]



PIN	CN17			CN19[100ohm]
3-2	P1_MDIP	[U98]88Q2112 MDIP	[U106]88Q5040 P1_MDIP	P1_MDIP_100
3-1	P1_MDIN	[U98]88Q2112 MDIN	[U106]88Q5040 P1_MDIN	P1_MDIN_100
4-2	P2_MDIP	[U99]88Q2112 MDIP	[U106]88Q5040 P2_MDIP	P2_MDIP_100
4-1	P2_MDIN	[U99]88Q2112 MDIN	[U106]88Q5040 P2_MDIN	P2_MDIN_100
2-1	P3_MDIP	[U100]88Q2112 MDIP	[U106]88Q5040 P3_MDIP	P3_MDIP_100
2-2	P3_MDIN	[U100]88Q2112 MDIN	[U106]88Q5040 P3_MDIN	P3_MDIN_100
1-1	P4_MDIP	[U101]88Q2112 MDIP	[U106]88Q5040 P4_MDIP	P4_MDIP_100
1-2	P4_MDIN	[U101]88Q2112 MDIN	[U106]88Q5040 P4_MDIN	P4_MDIN_100

4.9. CN18

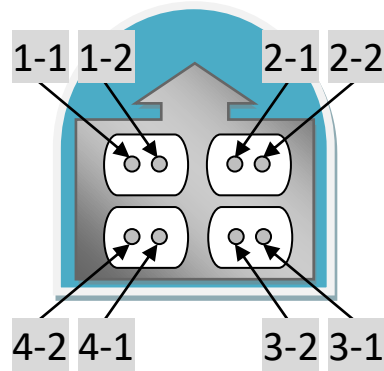
[410AF08C6TN1T1CZ]



PIN	CN18	
3-2	P5_MDIP	[U102]88Q2112 MDIP
3-1	P5_MDIN	[U102]88Q2112 MDIN
4-2	P6_MDIP	[U103]88Q2112 MDIP
4-1	P6_MDIN	[U103]88Q2112 MDIN
2-1	P7_MDIP	[U104]88Q2112 MDIP
2-2	P7_MDIN	[U104]88Q2112 MDIN
1-1	P8_MDIP	[U105]88Q2112 MDIP
1-2	P8_MDIN	[U105]88Q2112 MDIN

4.10. CN19

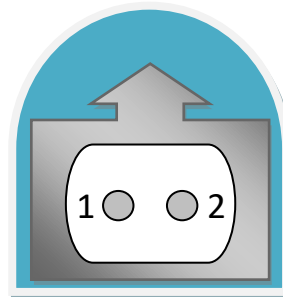
[410AF08C6TN1T1CZ]



PIN	CN19[100ohm]	CN17
3-2	P1_MDIP_100	P1_MDIP
3-1	P1_MDIN_100	P1_MDIN
4-2	P2_MDIP_100	P2_MDIP
4-1	P2_MDIN_100	P2_MDIN
2-1	P3_MDIP_100	P3_MDIP
2-2	P3_MDIN_100	P3_MDIN
1-1	P4_MDIP_100	P4_MDIP
1-2	P4_MDIN_100	P4_MDIN

4.11. CN22

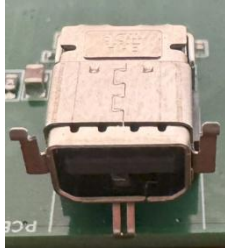
[5440101001Z] Ethernet TSN T1 PHY



PIN	CN22	Ethernet TSN T1 PHY
1	1	[U113]88Q2112 MDIP
2	2	[U113]88Q2112 MDIN

4.12. CN8

[Mini DisplayPort Connector]



4.13. CN11

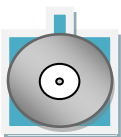
[DSI0] FAKRA to Display



PIN	CN11	[U28]MAX96789
1	GMSL_DISPLAY_B	DSI0 SIOBP

4.14. CN10

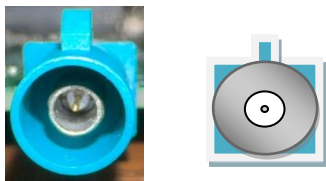
[DSI0] FAKRA to Display



PIN	CN10	[U28]MAX96789
1	GMSL_DISPLAY_A	DSI0 SIOAP

4.15. CN38

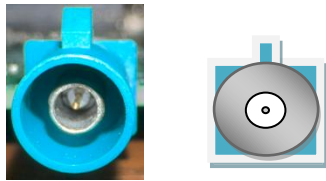
[DSI1] FAKRA to Display



PIN	CN38	[U181]MAC96789
1	GMSL_DISPLAY_B	DSI1 SIOBP

4.16. CN37

[DSI1] FAKRA to Display



PIN	CN38	[U181]MAC96789
1	GMSL_DISPLAY_A	DSI1 SIOAP

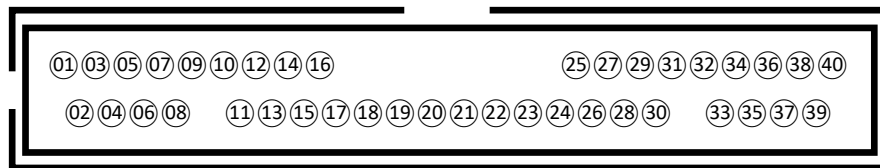
4.17. CN9

[HDMI]



4.18. CN32

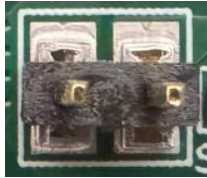
[External Connector] CAN/I2C/GPIO CONNECTOR



Function	PIN	PIN	Function
I2C2_SDA_CNA	1	2	I2C2_SCL_CNA
I2C2_SDA_CNB	3	4	I2C2_SCL_CNB
I2C2_SDA_CNC	5	6	I2C2_SCL_CNC
I2C2_SDA_CND	7	8	I2C2_SCL_CND
I2C2_SDA_CNE	9	10	I2C2_SCL_CNE
I2C2_SDA_CNF	11	12	I2C2_SCL_CNF
D3.3_I2C	13	14	SG
IRQ2/MD0_SW	15	16	GP8_12_V
GP8_13/MD27_SW	17	18	GP6_05/AVB1_3V3
D5.0V1	19	20	D5.0V1
CANL0_CN	21	22	CANH0_CN
CANL1_CN	23	24	CANH1_CN
CANL2_CN	25	26	CANH2_CN
CANL3_CN	27	28	CANH3_CN
CANL4_CN	29	30	CANH4_CN
CANL5_CN	31	32	CANH5_CN
CANL6_CN	33	34	CANH6_CN
CANL7_CN	35	36	CANH7_CN
BLP_A_CN	37	38	BLM_A_CN
BLP_B_CN	39	40	BLM_B_CN
SG	H1	H4	SG
SG	H2	H5	SG
SG	H3	H6	SG

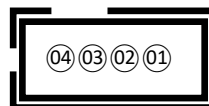
4.19. JP1

[EX_PWRON] Pin Header for external power control



4.20. CN33

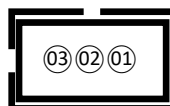
[FAN]



PIN	Function
1	GND
2	D12.0V1
3	FAN_SENSE
4	FAN_PWMCNT

4.21. CN34

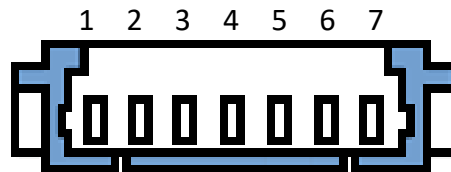
[FAN]



PIN	Function
1	PWM_OUT
2	D5.0V1_VSYS
3	NC

4.22. CN5

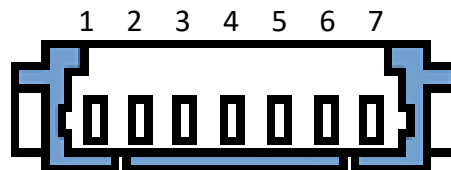
[SPI header]



PIN	Function
1	MFP9_RO_0A
2	MFP8_BNE_0A
3	MFP5_SCK_0A
4	CP3
5	MFP7_MISO_0A
6	MFP6_MOSI_0A
7	GND

4.23. CN7

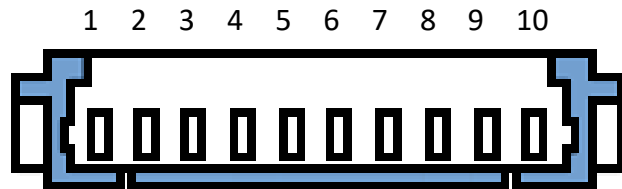
[SPI header]



PIN	Function
1	MFP9_RO_0B
2	MFP8_BNE_0B
3	MFP5_SCK_0B
4	CP6
5	MFP7_MISO_0B
6	MFP6_MOSI_0B
7	GND

4.24. CN20

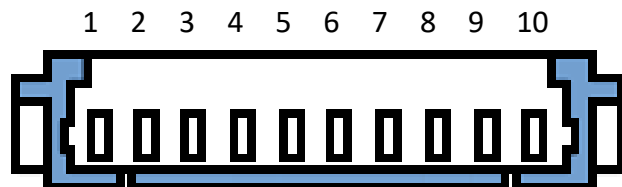
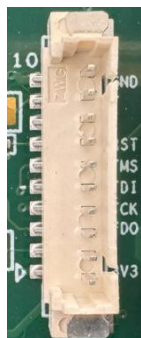
Connect to [U106]88Q5040



PIN	Function
1	88Q5040_3V3
2	TP26
3	BONSAI_TDO
4	BONSAI_TCK
5	BONSAI_TDI
6	BONSAI_TMS
7	BONSAI_RSTn
8	NC
9	TP27
10	GND

4.25. CN16

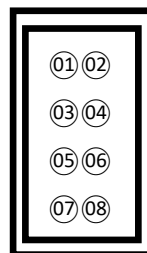
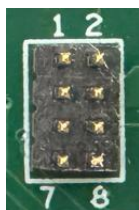
Connect to [U88]88Q6113



PIN	Function
1	Q6113_3V3_VDDO
2	TP17
3	TDO_ES
4	TCK_ES
5	TDI_ES
6	TMS_ES
7	TRSTN_ES
8	NC
9	TP18
10	GND

4.26. CN3

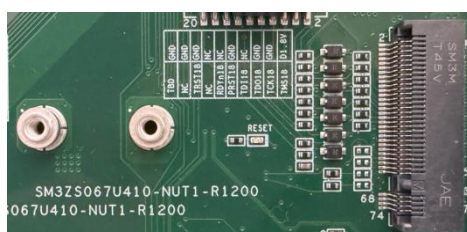
[JTAG2]



Function	PIN	PIN	Function
TDO2	1	2	D3.3V
TDI2	3	4	D1.8V
TMS2	5	6	PRESETN_18
TCK2	7	8	GND

4.27. CN14

[2242 Key M for HSSTP/Hailo-8L] [2230 Key M M.2 for SSD]



Function	PIN	PIN	Function
----------	-----	-----	----------

CN_CONFIG3	1	2	D3.3V1_M2
GND	3	4	D3.3V1_M2
PCIE1_RX1_M_RE_CN	5	6	NC
PCIE1_RX1_P_RE_CN	7	8	NC
GND	9	10	NC
PCIE1_TX1_M_CN_RE	11	12	D3.3V1_M2
PCIE1_TX1_P_CN_RE	13	14	D3.3V1_M2
GND	15	16	D3.3V1_M2
14 PCIE1_RX0_M_RE_CN	17	18	D3.3V1_M2
PCIE1_RX0_P_RE_CN	19	20	NC
CN_CONFIG0	21	22	D1.8V
PCIE1_TX0_M_CN_RE	23	24	NC
PCIE1_TX0_P_CN_RE	25	26	TCK_18
GND	27	28	TMS_18
PCIE0_RX1_M_RE	29	30	TRSTn_18
PCIE0_RX1_P_RE	31	32	TDI_18
GND	33	34	TDO_18
PCIE0_TX1_M_RE	35	36	PRESETN_18
PCIE0_TX1_P_RE	37	38	NC
GND	39	40	NC
PCIE0_RX0_M_RE	41	42	NC
PCIE0_RX0_P_RE	43	44	NC
GND	45	46	NC
PCIE0_TX0_M_RE	47	48	NC
PCIE0_TX0_P_RE	49	50	HSSTP_RESET
GND	51	52	(NI)P3V3_4
PCIE0_CLK_M_CN	53	54	(NI)P3V3_4
PCIE0_CLK_P_CN	55	56	(NI) I2C2_SDA_M2M2
GND	57	58	(NI) I2C2_SCL_M2M2
GND	67	68	D3.3V1_M2
CN_CONFIG1	69	70	Short to pin [72] [74]
GND	71	72	Short to pin [70] [74]
GND	73	74	Short to pin [70] [72]
CN_CONFIG2	75		
GND	G1	G2	GND
NC	N1	N2	NC

4.28. CN13

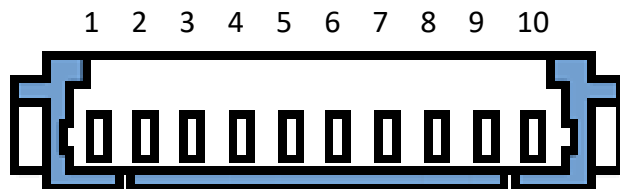
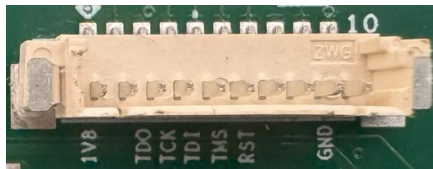
[OcuLink]



Function	PIN	PIN	Function
P3.3V	A1	B1	(NI) D5.0V1_OCU
GND	A2	B2	GND
PCIE_D_P10_PERP	A3	B3	PCIE_D_P10_PETP
PCIE_D_P10_PERN	A4	B4	PCIE_D_P10_PETN
GND	A5	B5	GND
PCIE_D_P11_PERP	A6	B6	PCIE_D_P11_PETP
PCIE_D_P11_PERN	A7	B7	PCIE_D_P11_PETN
GND	A8	B8	GND
PCIE_CN7_A9_SCL	A9	B9	PCIE_CN7_B9_BP_TYPE
PCIE_CN7_A10_SDA	A10	B10	PCIE_CN7_B10_CWAKEn
GND	A11	B11	GND
PCIE_CN7_A12_PERSTn	A12	B12	PCIE_CN7_B12_VSP_M
PCIE_CN7_A13_CPRSNTn	A13	B13	PCIE_CN7_B13_VSP_P
GND	A14	B14	GND
NC	A15	B15	NC
NC	A16	B16	NC
GND	A17	B17	GND
NC	A18	B18	NC
NC	A19	B19	NC
GND	A20	B20	GND
(NI) D5.0V1_OCU	A21	B21	P3.3V

4.29. CN12

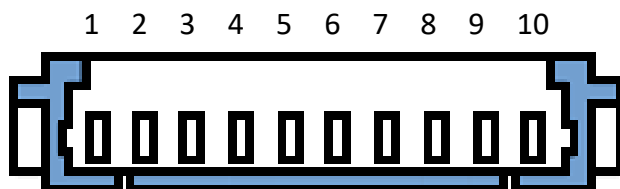
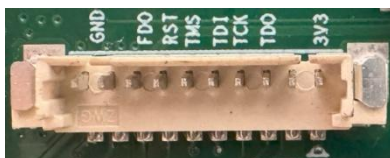
Connect to [U51] PCIe switch



PIN	Function
1	P1.8V
2	TP4
3	PCIE_TDO
4	PCIE_TCK
5	PCIE_TDI
6	PCIE_TMS
7	PCIE_RST
8	NC
9	TP5
10	GND

CN36

Connect to [U172] MCU JTAG

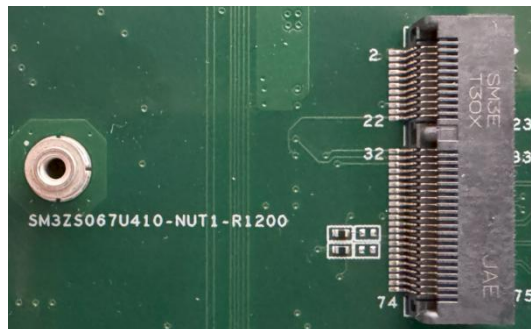


PIN	Function
1	MCU_3V3
2	TP31
3	JTAG-DCUTDO
4	JTAG-DCUTCK

5	JTAG-DCUTDI
6	JTAG-DCURDY
7	JTAG-DCURST
8	JTAG-DCUTMS
9	TP32
10	GND

4.30. J1

[2230 Key E M.2 for WIFI/BT] PCIe SW Port2

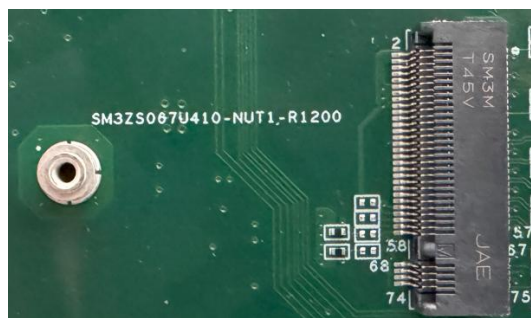


Function	PIN	PIN	Function
GND	1	2	D3.3V1_M2
NC	3	4	D3.3V1_M2
NC	5	6	NC
GND	7	8	NC
NC	9	10	NC
NC	11	12	NC
NC	13	14	NC
NC	15	16	NC
NC	17	18	GND
NC	19	20	NC
NC	21	22	M2_UART_RXD
NC	23		
		32	M2_UART_TXD
GND	33	34	M2_UART_CTS

PCIE_D_P20_PETP	35	36	M2_UART_RTS
PCIE_D_P20_PETN	37	38	NC
GND	39	40	NC
PCIE_D_P20_PERP	41	42	NC
PCIE_D_P20_PERN	43	44	NC
GND	45	46	NC
REFCLKP_SL2	47	48	NC
REFCLKN_SL2	49	50	NC
GND	51	52	PERST#_SL2_L
P3V3_2	53	54	NC
P3V3_2	55	56	NC
GND	57	58	(NI) I2C2_SDA_M2E
PCIE_D_P21_PETP	59	60	(NI) I2C2_SCL_M2E
PCIE_D_P21_PETN	61	62	NC
GND	63	64	NC
PCIE_D_P21_PERP	65	66	NC
PCIE_D_P21_PERN	67	68	NC
GND	69	70	NC
NC	71	72	D3.3V1_M2
NC	73	74	D3.3V1_M2
GND	75		
GND	G1	G2	GND
NC	N1	N2	NC

4.31. CN15

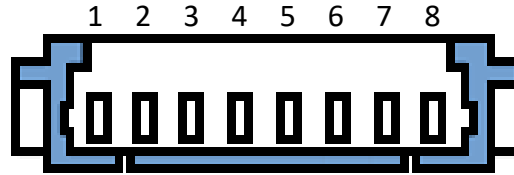
[2230 Key M M.2 for SSD] PCIe SW Port3



Function	PIN	PIN	Function
P3_CONFIG3	1	2	D3.3V1_M2
GND	3	4	D3.3V1_M2
NC	5	6	NC
NC	7	8	NC
GND	9	10	NC
NC	11	12	D3.3V1_M2
NC	13	14	D3.3V1_M2
GND	15	16	D3.3V1_M2
NC	17	18	D3.3V1_M2
NC	19	20	NC
P3_CONFIG0	21	22	NC
NC	23	24	NC
NC	25	26	NC
GND	27	28	NC
PCIE_D_P31_PERN	29	30	NC
PCIE_D_P31_PERP	31	32	NC
GND	33	34	NC
PCIE_D_P31_PETN	35	36	NC
PCIE_D_P31_PETP	37	38	NC
GND	39	40	NC
PCIE_D_P30_PERN	41	42	NC
PCIE_D_P30_PERP	43	44	NC
GND	45	46	NC
PCIE_D_P30_PETN	47	48	NC
PCIE_D_P30_PETP	49	50	PERST#_SL3_L
GND	51	52	(NI) P3V3_3
REFCLKN_SL3	53	54	(NI) P3V3_3
REFCLKP_SL3	55	56	(NI) I2C2_SDA_M2M1 12
GND	57	58	(NI) I2C2_SCL_M2M1
GND	67	68	D3.3V1_M2
P3_CONFIG1	69	70	Short to pin [72] [74]
GND	71	72	Short to pin [70] [74]
GND	73	74	Short to pin [70] [72]
P3_CONFIG2	75		
GND	G1	G2	GND
NC	N1	N2	NC

4.32. CN28

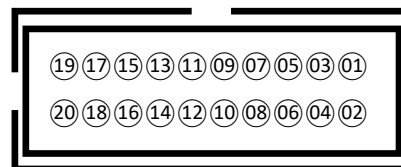
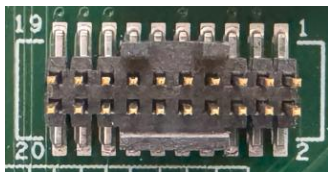
[I2S/SPI PIN HEADER]



PIN	Function
1	AUDIO_CLKOUT_V
2	GND
3	AUDIO_CLKIN_V
4	GND
5	SSI_SCK_V
6	SSI_WS_V
7	SSI_SD_V
8	D3.3V_SPI

4.33. CN2

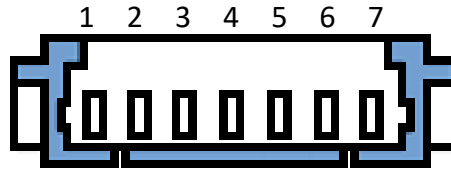
[JTAG1]



Function	PIN	PIN	Function
D1.8V	1	2	TMS_18
GND	3	4	TCK_18
GND	5	6	TDO_18
NC	7	8	TDI_18
GND	9	10	PRESETN_18
NC	11	12	RDYn_18
NC	13	14	NC
GND	15	16	TRSTn_18
GND	17	18	NC
GND	19	20	GND

4.34. CN31

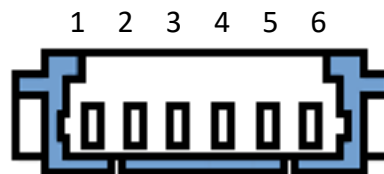
[I2S/SPI PIN HEADER]



PIN	Function
1	MSIOF1_SS2/MD18_SW
2	MSIOF1_SS1/MD16_SW
3	MSIOF1_SCK/MD14_SW
4	MSIOF1_SYNC/MD32_SW
5	MSIOF1_RXD/MD11_SW
6	MSIOF1_TXD/MD13_SW
7	GND

4.35. CN30

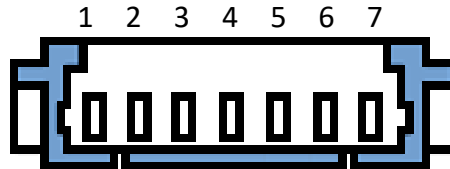
[GNSS(UART) PIN HEADER]



PIN	Function
1	D3.3V_SP
2	GP1_27/HSCIF3_HCTS3N_V
3	GP1_26/HSCIF3_HRTS3N_V
4	GND
5	GP1_24/SL_SW1_V
6	GP1_28/HSCIF3_HTX3_V

4.36. CN27

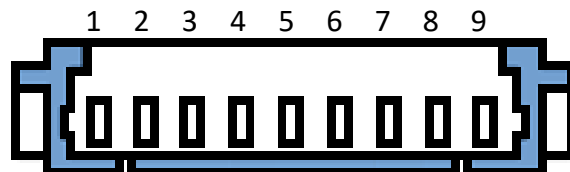
[I2S/SPI PIN HEADER]



PIN	Function
1	MSIOF2_SS2/MD25_SW
2	MSIOF2_SS1/MD24_SW
3	MSIOF2_SCK/MD21_SW
4	MSIOF2_SYNC/MD23_SW
5	MSIOF2_RXD/MD19_V
6	MSIOF2_TXD/MD20_SW
7	GND

4.37. CN24

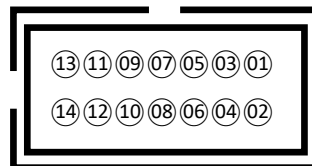
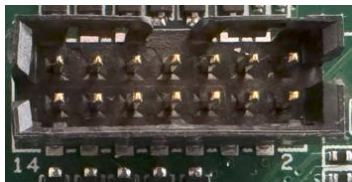
[DigRF CN]



PIN	Function
1	IPC_RX_N
2	IPC_RX_P
3	GND
4	IPC_TX_P
5	IPC_TX_N
6	GND
7	IPC_CLKIN_18
8	GND
9	IPC_CLKOUT_18

4.38. CN25

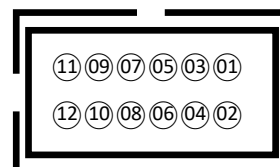
[I2C Connector]



Function	PIN	PIN	Function
D3.3V_I2C	1	2	GND
D3.3V_I2C	3	4	GND
I2C0_SCL	5	6	I2C0_SDA
I2C1_SCL	7	8	I2C1_SDA
I2C2_SCL	9	10	I2C2_SDA
I2C3_SCL	11	12	I2C3_SDA
I2C4_SCL	13	14	I2C4_SDA

4.39. CN29

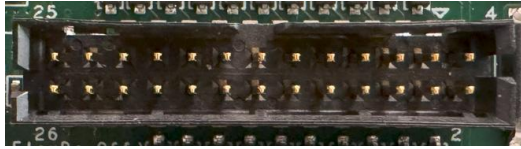
[GPIO Connector]



Function	PIN	PIN	Function
D3.3V_SPI	1	2	IRQ2/MD0_SW
GND	3	4	GND
D3.3V_I2C	5	6	GP8_12_V
GND	7	8	GP8_13/MD27_SW
D3.3V	9	10	NC
GND	11	12	GP6_05/AVB1_3V3

4.40. CN26

[CAN BOARD Connector]



25	23	21	19	17	15	13	11	09	07	05	03	01
26	24	22	20	18	16	14	12	10	08	06	04	02

Function	PIN	PIN	Function
D12.0V2	1	2	D5.0V1
VBOOST	3	4	CAN_D3.3V
CANFD0_RX_D	5	6	CANFD4_RX_D
CANFD0_TX_D	7	8	CANFD4_TX_D
CANFD1_RX_D	9	10	CANFD5_RX_D
CANFD1_TX_D	11	12	CANFD5_TX_D
CANFD2_RX_D	13	14	CANFD6_RX_D
CANFD2_TX_D	15	16	CANFD6_TX_D
CANFD3_RX_D	17	18	CANFD7_RX_D
CANFD3_TX_D	19	20	CANFD7_TX_D
CAN_SILENT	21	22	GND
GND	23	24	GND
GND	25	26	GND

5. APPENDIX